PROJECT TITLE - Eazydiner analysis using EDA and Webscrapping

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ABOUT ME -

- **Name :-** Disha Sudhakar Surywanshi
- **Education** :- G. H. Raisoni college of engineering and business management ,Jalgaon.
- **Why I Want To Learn Data Science** :- I want to learn data science because ,I've always been curious about how companies makes the decision about what customers likes or dislikes ,that urge of knowing how this decision making works in companies created my interest in data science .
- **Connect With Me :-**
- **LinkedIn -** www.linkedin.com/in/dishasuryawanshi0104
- **GitHub -** https://github.com/Dishasuryawanshi0104

BUSINESS OBJECTIVES

- Analyse how restaurants are distributed across cities, locations, cuisines, and meal types.
- Identify price ranges, affordability, and whether premium locations charge higher prices.
- Understand how ratings vary with price, cuisine, meal type, and location.
- Examine how offers/discount percentages vary by price range, restaurant type, and rating.
- Detect restaurants that offer high ratings at relatively lower prices, helping customers make better choices.

WEB-SCRAPPING : DETAILS

- Data was collected from the EazyDiner website.
- Restaurants information was extracted using web scraping.
- The dataset represents real-time publicly available restaurant listings.
- Scraped attributes include:
  - Restaurant Name
  - City & Location
  - Cuisine
  - Meal Type
  - Price
  - Ratings
  - Offers

DATA SUMMARY

Database Schema

- **Data Source:** EazyDiner website (web scraped)
- **Data Format:** Structured tabular data (CSV/DataFrame)

Total Records	5692
Total Features	9(Columns)
Cities Covered	10
Unique Cuisines	500
Price Range	₹150 - ₹8000

Data Quality & Readiness

- Missing values and inconsistent formats due to web scraping
- Duplicates identified and removed
- Data cleaned and standardised for EDA

DATA CLEANING

Missing Value Handling

- Identified missing values in price, rating, and offers
- Applied appropriate removal or imputation techniques.
- City has missing values because some restaurants don't have a city in data

Data Type Standardisation

- Converted offers and prices from text to numeric format
- Removed currency symbols and special characters

```
hotel           0
address         0
ratings         3
prices          0
cuisines       21
offers          0
Meal_Type       0
city         1590
dtype: int64
```

Missing Values

```
Restaurants                        0
address                            0
city                               0
ratings                            0
prices                             0
cuisines                           0
Meal_Type                          0
Restaurant_offer_percentage        0
Online_payment_offer_percentage    0
dtype: int64
```

After handling missing values

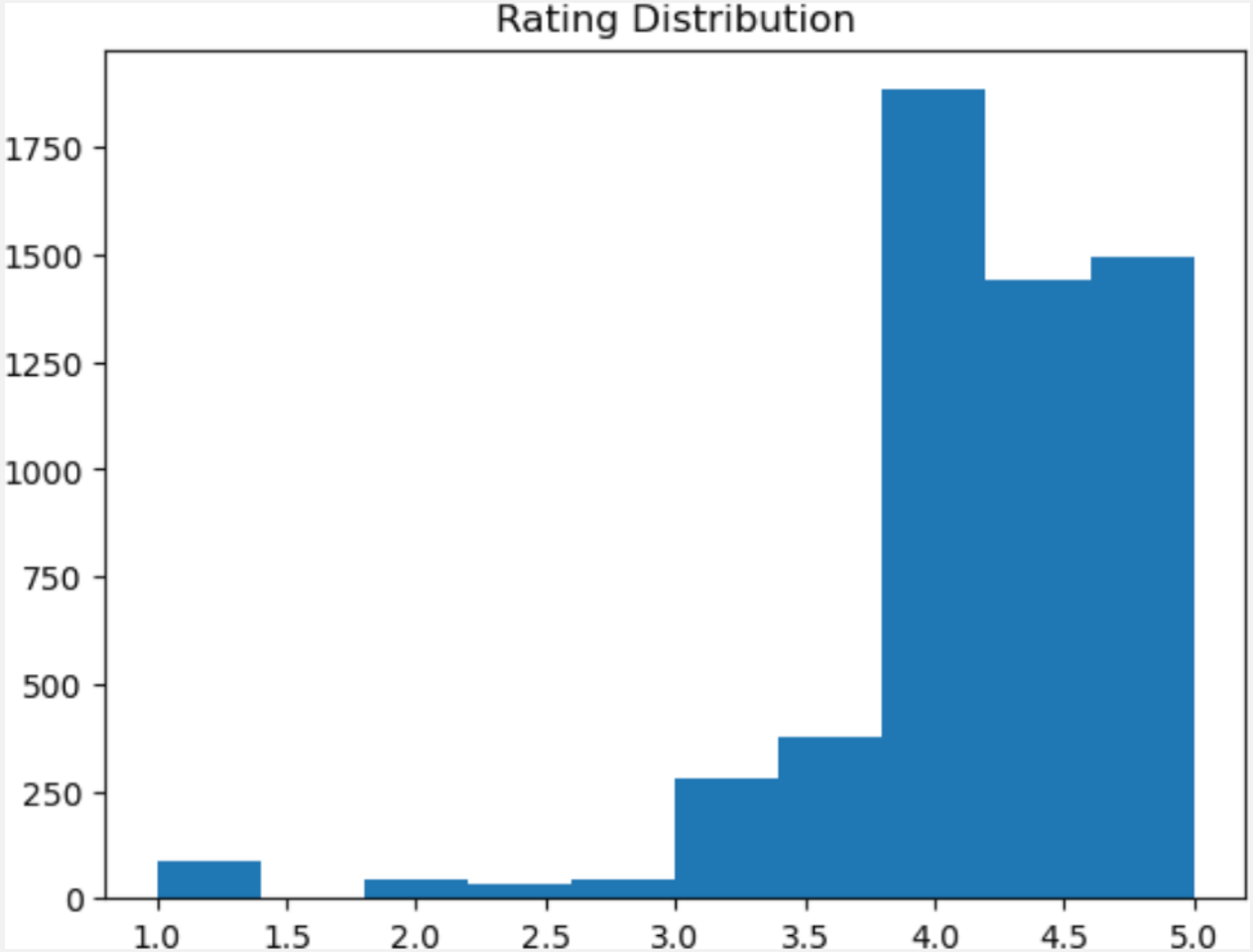
```
hotel          object
address        object
ratings       float64
prices         object
cuisines       object
offers         object
Meal_Type      object
dtype: object
```

Before Converting data types

```
Restaurants                        object
address                            object
city                               object
ratings                           float64
prices                              int64
cuisines                           object
Meal_Type                          object
Restaurant_offer_percentage         int64
Online_payment_offer_percentage     int64
dtype: object
```

After Converting data types

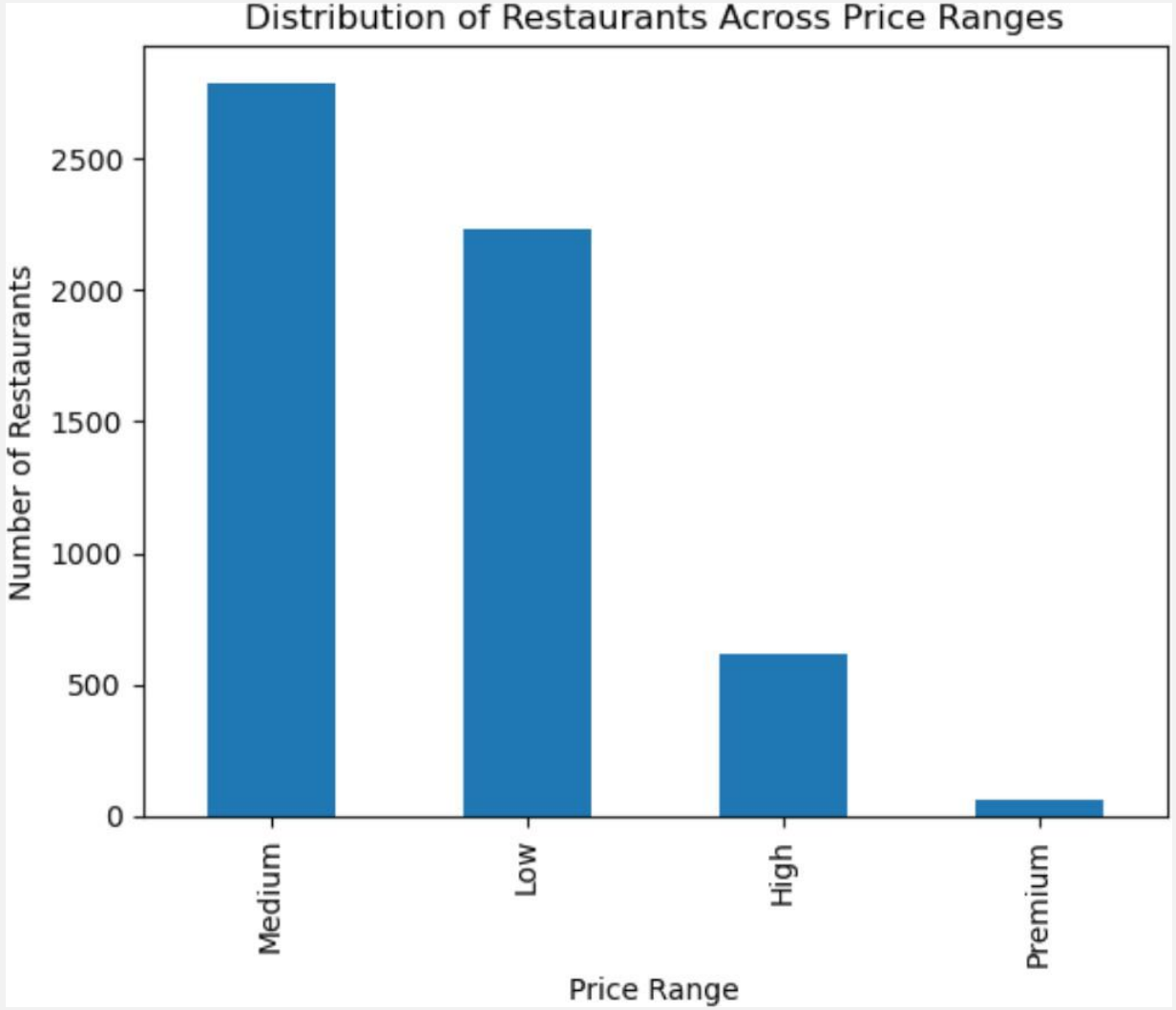
RATINGS DISTRIBUTION



Insights :-

- The highest bars are in the 4.0–4.5 range
- This shows most restaurants are well-rated
- Indicates generally positive customer satisfaction.

DISTRIBUTION OF RESTAURANTS ACROSS PRICE RANGES



Insights :-

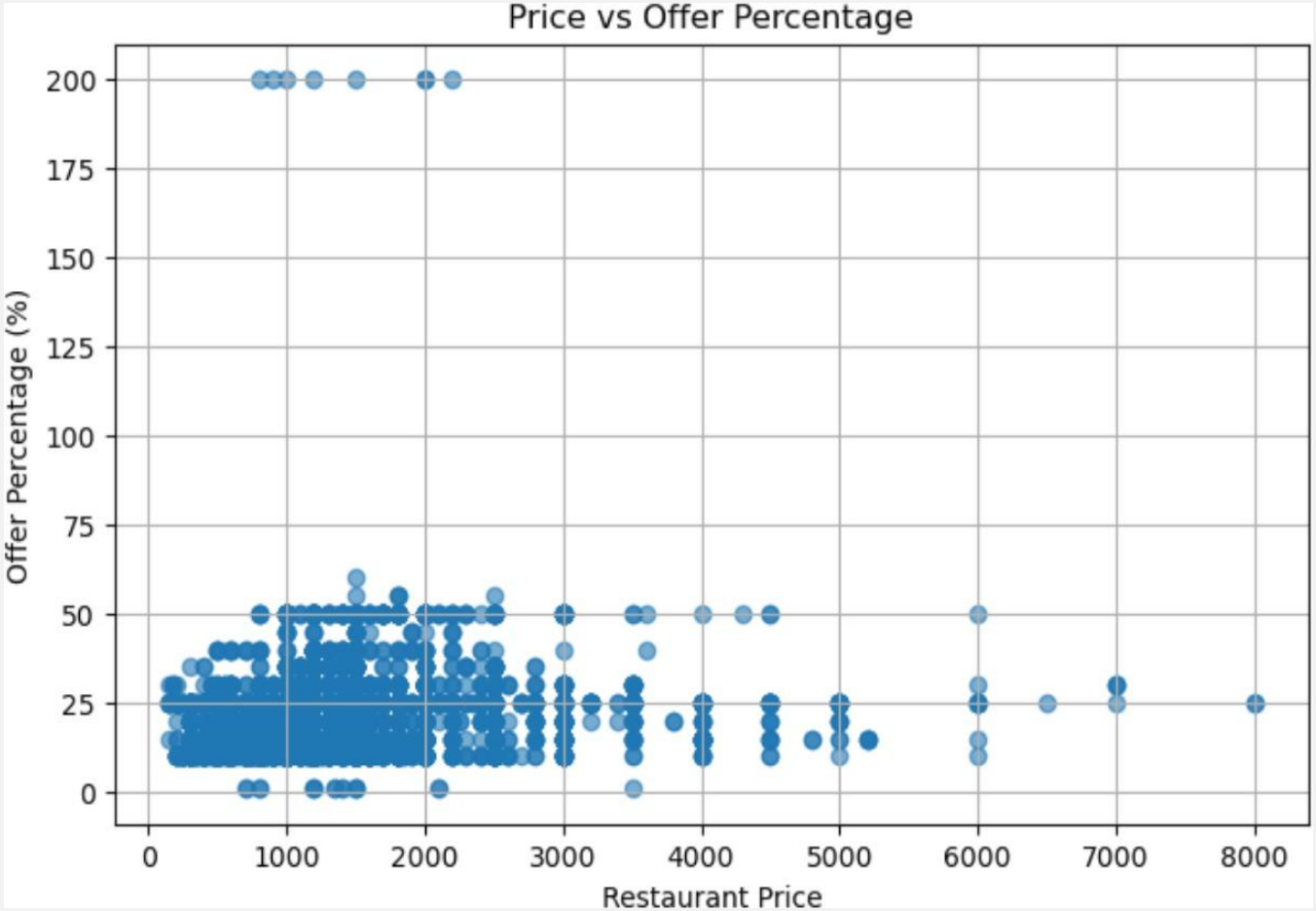
- The platform is largely dominated by affordable to mid-range dining options, targeting mass-market customers.
- Fewer restaurants operate in the high-price segment, possibly due to limited demand or higher operational costs.

RELATIONSHIP BETWEEN PRICE AND RATING

Insights :-

- Restaurants across all price ranges receive
- both high and low ratings
- High ratings (4 - 5) are not exclusive to expensive restaurants.
- Affordable restaurants dominate the dataset and show high variability in customer satisfaction.

RELATIONSHIP BETWEEN RESTAURANT PRICE AND OFFER PERCENTAGE



Insights :-

- Discount strategies are mainly used to attract customers in budget and mid-range restaurants.
- Premium restaurants rely more on brand value and experience rather than heavy discounts.

CHALLENGES

- EazyDiner uses dynamically loaded content, making scraping more complex and time-consuming
- Columns like ratings and offers contained both text and numeric values, complicating preprocessing.
- Some restaurants had missing, ratings, city names, requiring extensive cleaning.

CONCLUSIONS AND RECOMMENDATIONS

Conclusions :-

- Market is dominated by mid-range restaurants.
- High ratings are common, but mostly clustered Ratings are heavily concentrated between 3.5 and 4.5.
- Discounts are mainly used by budget and mid-range restaurants.
- Customers prioritise quality and experience over discounts
- Offers are more of a marketing tool than a quality indicator

Recommendations :-

- Restaurants should focus on food quality and service, as higher prices do not guarantee higher ratings.
- Mid-range restaurants should use moderate discounts strategically to attract price-sensitive customers.
- Premium restaurants should avoid heavy discounting and instead highlight experience, ambience, and exclusivity.
- Budget restaurants can leverage offers as a customer acquisition tool in competitive markets.

THANK YOU